

Braedyn Thompson

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EDUCATION

University of California, Berkeley

Graduation: Fall 2027

Major: Data Science & Computer Science - Applied Mathematics & Modeling Emphasis

GPA: 3.75/4.00

Coursework: Artificial Intelligence, Data Structures, Principles & Techniques of Data Science, Data Engineering, Data Mining and Analytics, Efficient Algorithms

WORK & PROFESSIONAL EXPERIENCE

Lawrence Berkeley National Laboratory

Berkeley, CA

Data Science & Machine Learning Research Intern

Sep 2026 – Present

- Implemented PatchTST (PyTorch/neuralforecast) for multivariate time-series forecasting of cache-traffic signals across multiple horizons, tuning lookback windows and beating a persistence baseline on all targets by up to 24% RMSE.
- Exposed the ceiling on peak-load predictability and guided feature/model design by building reproducible Pandas EDA pipelines over 3 years of XCache traffic logs, visualizing seasonality, change-points, and right-skewed peaks with Matplotlib, Seaborn, and Plotly, and using rolling cross-validation to show models under-predict rare peaks.

California Baptist University - SeismicSoCal.Duckdns.org

Riverside, CA

Machine Learning Research Intern

May 2026 – Present

- Diagnosed the root cause of a stalled inverse-FEA predictor by benchmarking 7 ML models (Random Forest, XGBoost, HGB/ExtraTrees ensembles, PCA pipelines) and running paired clean-vs-noisy signal analysis, proving the $R^2 \approx 0.10$ ceiling was a data limitation rather than a model limitation and redirecting the team from futile model tuning.
- Raised earthquake-detection accuracy to 0.992 ROC-AUC (vs. 0.550 baseline) and magnitude-estimation R^2 to 0.840 by training PyTorch CNN→Transformer and GNN ensembles on 5,800+ multi-station SCEDC waveform windows.
- Built a real-time daemon streaming 10 SeedLink stations that runs continuous PyTorch CNN→Transformer detection plus GNN-fused multi-station magnitude estimation, confirming events via graded multi-station coincidence and geographic move-out checks before dispatching FCM push alerts, deployed on an Oracle Cloud VM under systemd.

NumIsToken

Berkeley, CA

Full-Stack Software Engineering Intern

Sep 2025 – Jun 2026

- Enabled resumable redemptions for previously unrecoverable transactions by building Redemption History and Detail services with robust state management and payment resumption logic, eliminating a class of stuck-transaction failures.
- Reduced redemption action load times by ~85% (~2–3s to ~300ms) by removing redundant function calls to streamline backend lifecycle APIs, then containerizing and deploying backend services with Docker for production readiness.

PROJECTS

Academy of Testers – academyoftesters.com

Berkeley, CA

Machine Learning & Artificial Intelligence Software Developer

Jan 2023 – Jun 2026

- Architected an adaptive learning engine that models per-student skill mastery by combining Bayesian Knowledge Tracing with Item Response Theory, selecting questions to maximize information gain at learner's estimated ability, and extended the 8-skill model with forgetting-curve decay and prerequisite-graph penalty propagation, powering a mastery radar.
- Improved AP essay-grading consistency, as measured by a ~23% reduction in scoring error (QWK 0.62) against official College Board scores, by building a RAG pipeline that embeds each free response, retrieves rubric clauses and score banded exemplars from a Postgres vector store using cosine similarity, and returns calibrated results as constrained JSON.

BearLM

Berkeley, CA

Data & Artificial Intelligence Engineer

Aug 2026 – Present

- Built a fully local, zero-cost RAG Q&A assistant (LangChain, Ollama, Llama 3.1 8B, nomic-embed-text, Chroma) over ~10K chunks spanning UC Berkeley CS/DS courses; blended keyword and meaning-based search (BM25 + vector embeddings) with reranking to pull the most relevant sources lifting recall@1 from 56% to 82% and recall@5 to 86%.
- Reduced hallucinated answers by grounding every response strictly in its retrieved sources, combining a relevance filter that scores and drops weak matches, query rewriting for sharper retrieval query, strict prompting answers only from context, and clickable citations page-level citations; raising RAGAS faithfulness from 0.55 to 0.83 (0.97 relevance).
- Engineered an ETL pipeline ingesting heterogeneous files (PDF, md, ipynb), token-chunking them with metadata, and batch-embedding ~10K records with retry/backoff into Chroma, BM25, and SQLite for hybrid search and analytics.

TECHNICAL SKILLS

Python, SQL, PostgreSQL, MySQL, SQLite, Pandas, NumPy, ETL Pipelines, Data Ingestion, Vector Search, Chroma, Hybrid Search BM25, Batch Embedding, FastAPI, Docker, Unit & Integration Testing, Matplotlib, Plotly, LangChain, RAG, RAGAS, Git